

Benchmarking and Performance Analysis of LLM Inferencing Systems

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Systems for LLMs
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1. Radix Tree Workload Modes

2. Finding the Sweet Zone



Full Sharing (All share 3L)

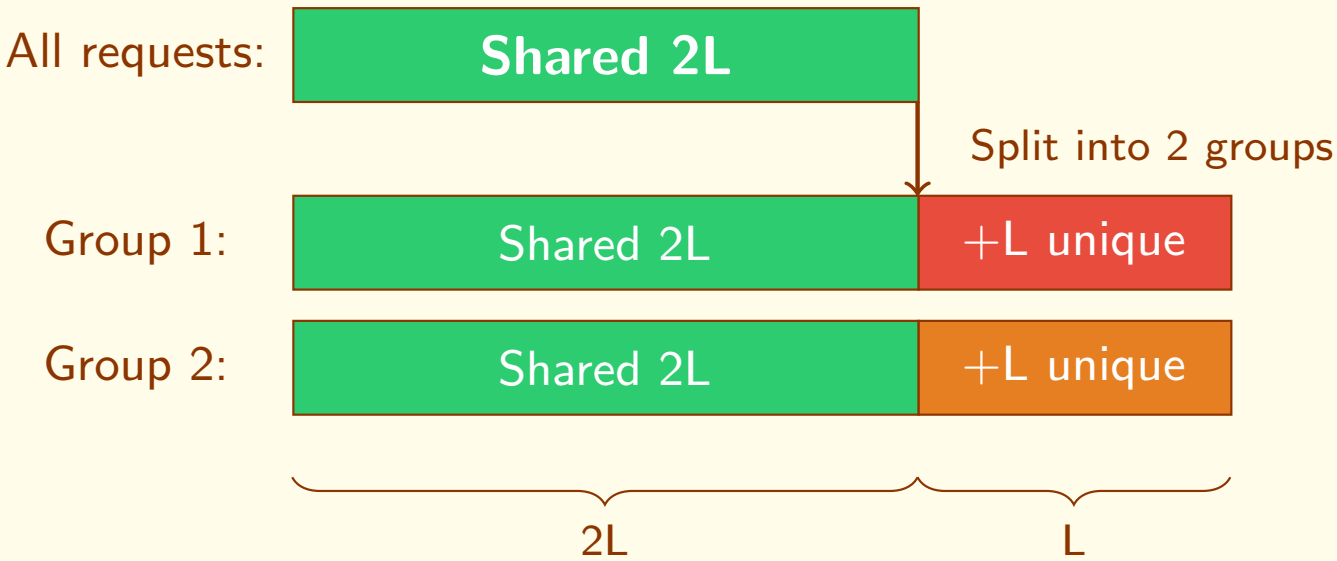
All requests: **Shared 3L**

3L tokens

Mode 2



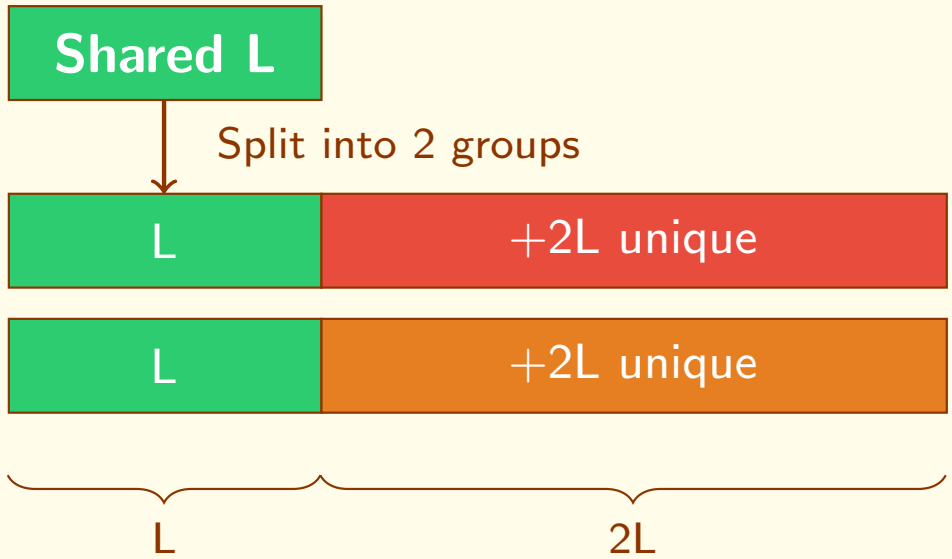
All share 2L, then 2xL





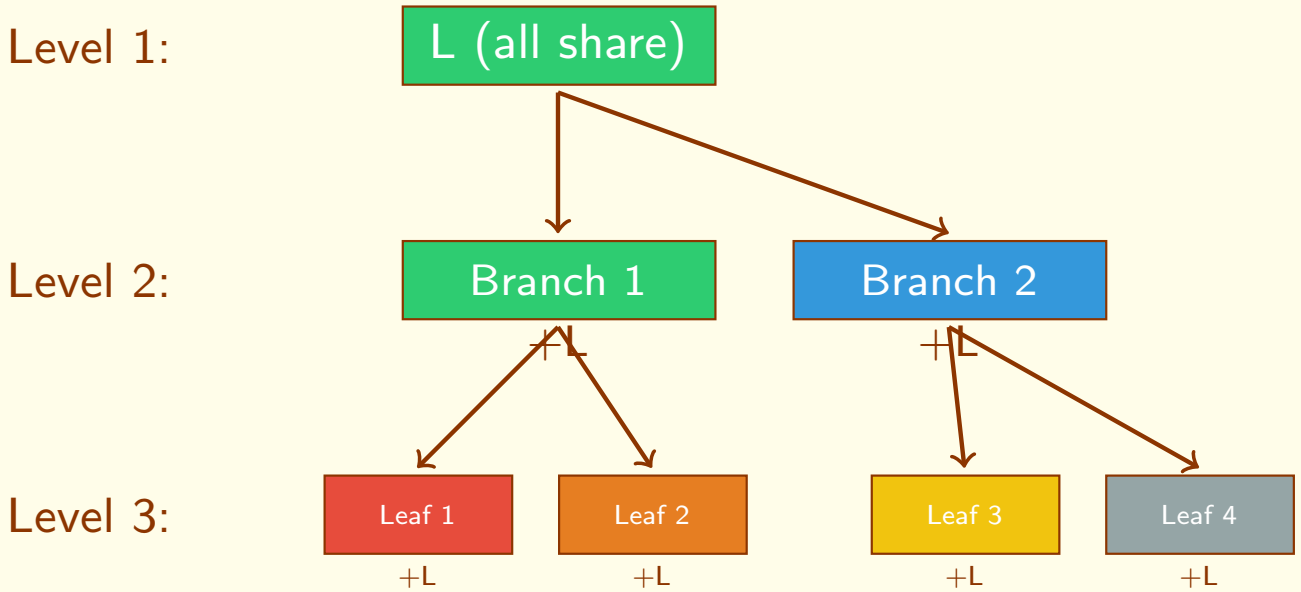
All share L, then 2x2L

All requests:





Three-Level Tree (L → 2L → 3L, 4 leaves)



Mode 5



2 separate groups with 3L each

Group 1: **Context A (3L tokens)**

Group 2: **Context B (3L tokens)**

3L tokens (completely different)

Results across varying L

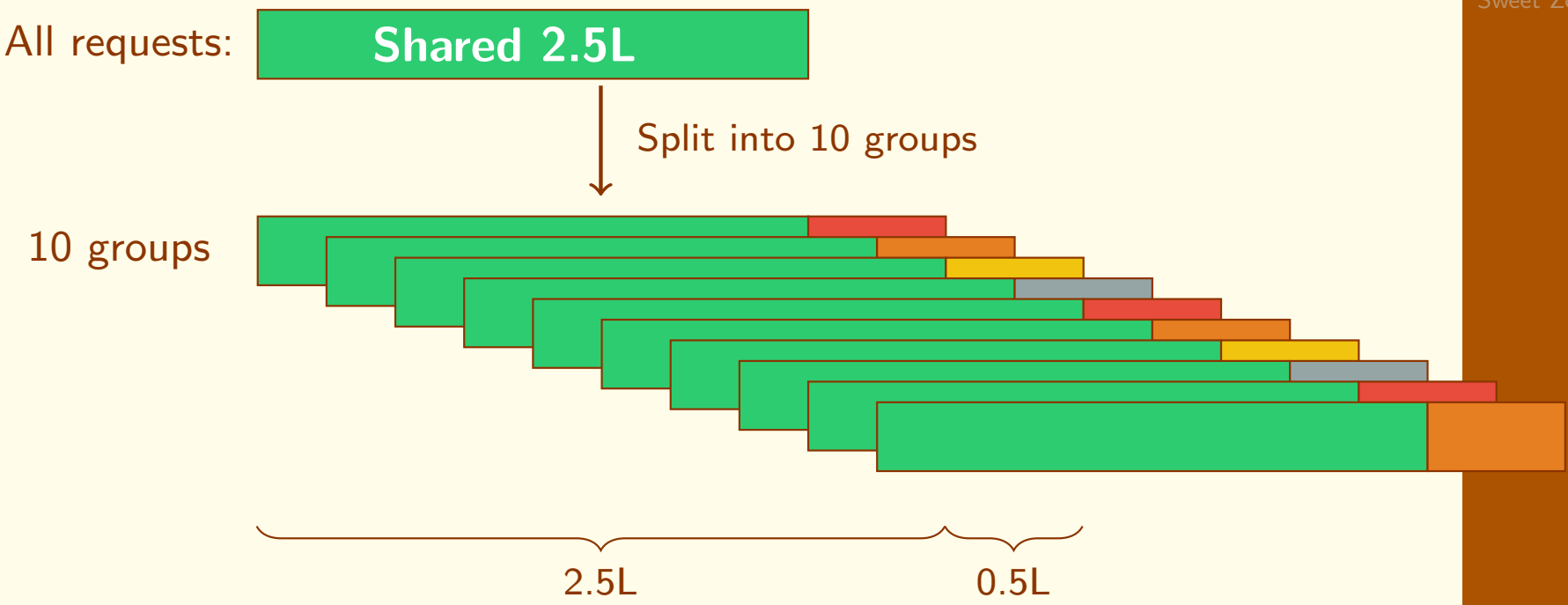


Figure: Results across varying L

Mode 6



2.5L shared + 10x0.5L



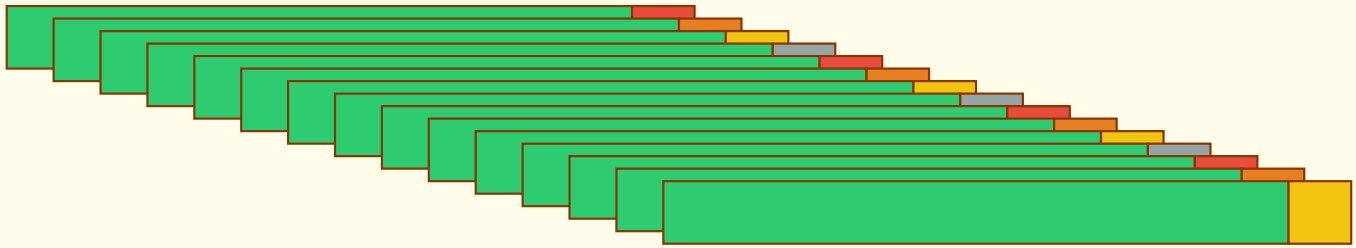
Mode 7



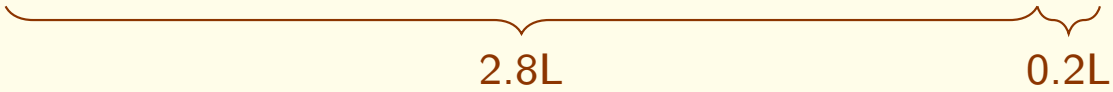
2.8L shared + 50x0.2L

All requests: **Shared 2.8L**

50 groups



...(More groups)



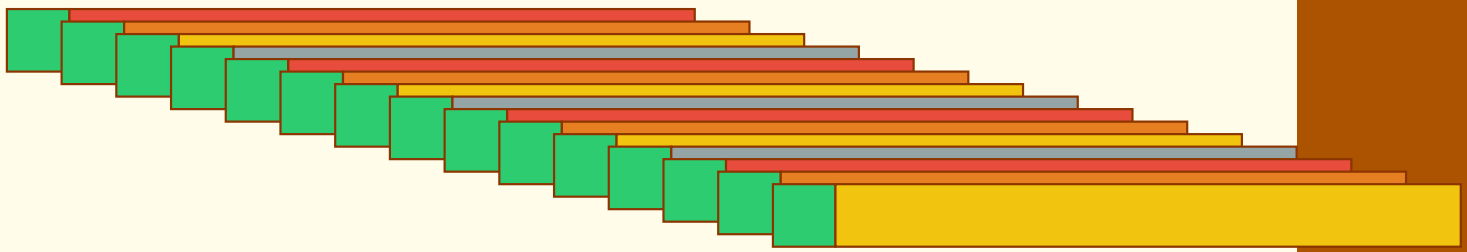
Mode 8



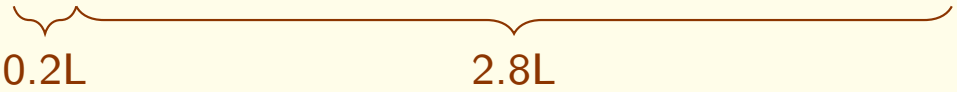
0.2L shared + 50x2.8L

All requests: 0.2L

50 groups



...(More groups)



Comparison of Throughput across different modes



Modes	L = 1000	L = 2000
Mode 1	29.18	27.97
Mode 2	19.94	14.06
Mode 3	15.40	8.89
Mode 4	15.41	9.88
Mode 5	10.18	6.62
Mode 6	25.34	20.27
Mode 7	28.05	23.17
Mode 8	10.05	6.48

Table: Comparison of Throughput across different modes

How many families?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- A shared prefix of length 6500 tokens
- 100 new output tokens
- Model: Llama3.1-8B

Number of Families	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
10	465.48	5.92	141.23	21.94
20	455.47	5.74	72.67	15.27
50	57.48	2.07	64.77	14.84
100	41.97	1.11	67.14	14.82

How many req/sec?



- 5 families
- Max batch size of 500
- A shared prefix of length 6500 tokens
- 100 new output tokens
- Model: Llama3.1-8B

req/sec	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
1	2.6	1.04	2.02	0.95
2	5.15	2.00	4.48	1.99
5	29.60	5.09	11.11	4.40
10	395.80	4.42	23.96	8.16
20	438.85	5.25	52.79	13.32

How long should the shared prefix be?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- 5 families
- 100 new output tokens
- Model: Llama3.1-8B

Shared Prefix Length in Number of Tokens	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
13	455.48	42.47	141.23	44.67
65	448.87	40.14	318.60	43.96
130	460.09	38.19	330.49	41.99
260	462.80	35.34	326.36	44.58
650	469.12	27.47	335.21	42.00

How large should the model be?



- A high request arrival rate (500 req/sec)
- Max batch size of 500
- 5 families
- 100 new output tokens
- Shared prefix length: 6500 tokens

Model Name	vLLM Baseline (FCFS)		vLLM (User-provided Family IDs)	
	Average Batch Size	Throughput	Average Batch Size	Throughput
Qwen2.5-0.5B	378.98	25.40	89.63	49.08
Llama3.1-8B	448.87	4.34	151.21	30.84
LongChat-13B	5.28	0.61	215.70	22.01



Thank You